



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical Electronics Engineering

Academic Year 2023 – 2024(Even Semester)

Degree, Semester & Branch: B.E/II/CIVIL

Course Code & Title: BE3252 BASIC ELECTRICAL, ELECTRONICS AND INSTRUMENTATION

Name of the Faculty member (s): Mrs V.Krishna Meera

Date: 06.03.24

Innovative Practice Description

- Unit / Topic: Unit I / Ohms law, KCL, KVL
- Course Outcome: CO1
- Topic Learning Outcome: TLO2.
- Activity Chosen: Ungraded Quiz
- Justification:

I explained the ohm's law, KCL, KVL. I solved some problems using ohm's law, KCL, KVL. After teaching this topic, I thought of this activity for making to have clarity about the ohm's law, KCL, KVL. So I can judge the level of understanding.

- Time Allotted for the Activity: 10 Minutes
- Details of the Implementation:
 - After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.
 - Total Strength :43
 - Reporter: Myself
 - At the end of the Class (Last 10 minutes)
 - I asked the students to think about the ohm's law, KCL, KVL.
 - Then I told them to write to attend the quiz which is ungraded(8 minutes)
 - Finally, I collected the responses from each students.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO5	PO8	PO10	PO12
C112.1	3	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO5	PO8	PO10	PO12
	3	2	1	1	1	1
Justification for correlation	Students will be solving basic problem on ohm's law, KCL and KVL	Suitable mathematical and engineering knowledge will be applied by the students to solve basic problems	Students will be using virtual laboratory for solving electric circuits.	The basic role of wind energy power generation in environmental sustainability is taught to the students	Students follow basic ethical practice to prepare the assignment	Presentation / Assignment will be submitted by the students on basic laws

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

I preferred this activity because it will help me to evaluate what types of questions I am asking students and what level they are working at.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students can understand their knowledge in that particular topic.

❖ *Challenges faced in implementation:*

References:

V. J. [Signature]
6/13/24.
Signature of Faculty Member


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Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: II Semester B.E CIVIL

Course Code & Title: BE3252 BASIC ELECTRICAL, ELECTRONICS AND
INSTRUMENTATION

Name of the Faculty member: Mrs.V.Krishna Meera, AP/ECE

Innovative Practice Description

❖ **Unit / Topic:** Unit III/ Electrical Wiring

• **Course Outcome:** CO 2

• **Topic Learning Outcome:** TLO 8

• **Activity Chosen:** Think-Pair-Share

• **Justification:**

- Electrical Wiring
- The topic Electrical Wiring and its types are used for the same purpose. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the different models concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

After teaching the concept, the students were made to pair with their neighbours.

Total Strength is 44, Number of Pairs –10

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about Electrical Wiring and its types for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the difference between the models for another 1 minute.
- ✓ Finally, I selected on Pair from each column randomly and ask them to share one difference between the concepts. (3 minutes)
- ✓ When each pair told the points simultaneously, I wrote the points on the board.
- ✓ Finally, I summarized the points and I told one missed point to all the students.
- ✓ With the help of PPT prepared (first slide- Activity name, Second slide-Topic for Think-pair-share, Third slide - answer) the answers were be projected to the students.
- ✓ After completion of the class I uploaded the PPT in the course website (Canvas)

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO10	PO12
C112.2	3	2	1	1	1

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO8	PO10	PO12
	3	2	1	1	1
Justification for correlation	Students will apply electronic engineering concepts to solve engineering problems. Hence it is substantially correlated (level 3).	Students will compare various embedded development strategies and contrast alternative solutions for firmware environments to select the best model. Hence, it is moderately correlated (level 2).	Students will be able to work with their team through Active Learning Methods. Hence it is slightly correlated (level 1).	Students will be able to communicate the technical concept through Active Learning Methods; hence it is slightly correlated (level 1).	Students will use the electronic fundamentals in the advancement of embedded system technology; hence it is slightly correlated (level 1).

• Images / Screenshot of the practice:





❖ *Reflective Critique:*

1. Pre-implementation Reflection :

I preferred this Activity because they know about wiring and different types of wiring .

Challenges anticipated:

- ❖ In the class mostly boys hastate to answer to the questions.
- ❖ Time utilization for conducting activity.

2. Post implementation Reflection:

• **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• **Challenges:**

- Slow learners were not able to understand some topics during discussion hours.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that one point was missed with respect to method of computation and finally I insisted the point they missed to tell.
- ✓ The success of the activity was evaluated by asking the same question in Internal
- ✓ Assessment test II – Around 70% of students answered correctly.

References:

1. <http://www.readwritethink.org/professional-development/strategy-guides/using-think-pair-share-30626.html>
2. <https://in.video.search.yahoo.com> – Think pair share

V. S. S. S.
24/3/21

Signature of Faculty Member

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Degree, Semester & Branch: II Semester B.E. CIVIL
Course Code & Title: BE3252 BASIC ELECTRICAL, ELECTRONICS AND
INSTRUMENTATION
Name of the Faculty member: Mrs.V.Krishna Meera, AP/ECE

Innovative Practice Description

- **Unit / Topic:** Unit III / Electrical Machines
- **Course Outcome:** CO 3/ Electrical Machines
- **Topic Learning Outcome:** TLO 11
- **Activity Chosen:** Demonstration
- **Justification:**
 - Explain Electrical Machines
 - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the DC Machines and AC Machines which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 43

Reporter: Myself

At the end the Class (Last 5 minutes)

- I asked the students to think about various types of meters in measurements concept for 2 minutes.
- Then I told them to pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.
- Finally, I have shown the DC Motor and AC Motor for each row (3 minutes)

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO8	PO10	PO12
C112.3	3	2	1	1	1	1

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO8	PO10	PO12
	3	2	1	1	1	1
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the working of meters.	Students will compare and contrast alternative solution processes to select the best process for the measurement.	Students will determine the design objectives and functional requirements of the instruments.	Students will follow the to prepare assignment on this topic..	Students will be able to create flow in a document or presentation.	Students will describe the rationale for the requirement for continuing professional development

• Images / Screenshot of the practice:



❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

• **Benefits:**

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

• **Challenges:**

- In the class mostly girls hesitate to answer the questions.
- Time utilization for conducting activity.

2. **Post-implementation Reflection :**

• **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• **Challenges:**

- Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

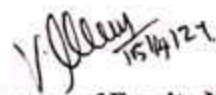
The assessment of effectiveness of the activity was felt when told most of the points.

➤ While conducting the activity, I understood that the students can able to explain the analog meters (Electrodynamometer and Wattmeter) construction and working principle.

➤ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 80%** of students answered.

References:

1. D P Kothari and I.J Nagarath, "Basic Electrical and Electronics Engineering", McGraw Hill Education (India) Private Limited, Second Edition, 2020
2. A.K. Sawhney, Puneet Sawhney 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2015.
3. S.K. Bhattacharya, Basic Electrical Engineering, Pearson Education, 2019 4. James A Svoboda, Richard C. Dorf, Dorf's Introduction to Electric Circuits, Wiley, 2018


Signature of Faculty Member


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Course Code & Title: BE3252 BASIC ELECTRICAL, ELECTRONICS AND INSTRUMENTATION

Name of the Faculty member (s): Mrs.V.Krishna Meera AP/ECE

Innovative Practice Description

- Unit / Topic: Unit IV / Applications of Diode

- Course Outcome: CO 4

- Topic Learning Outcome: TLO16.

- Activity Chosen: One Minute Paper

- Date :06.05.24

- Justification:

The structure, operation and V-I characteristics of PN junction diode. After teaching this topic ,I thought of this activity for making to have clarity about the operation of the diode and its characteristics. I can judge the understanding level of the students .

- Time Allotted for the Activity: 5 Minutes

- Details of the Implementation:

- After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.
- Total Strength :43
- Reporter: Myself
- At the end of the Class (Last 5 minutes)
- I asked the students to think about the diode operation and its applications concept for 3 minutes.
- Then I told them to write as much as they can within a short period of time (1 minute)
- Finally, I collected the papers from each column.

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO8	PO10	PO12
C112.4	3	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO8	PO10	PO12
	3	2	1	1	1	1
Justification for correlation	Students will be solving basic problem on Semiconductor devices using basic electrical concepts.	Suitable mathematical and engineering knowledge will be applied by the students to solve basic problems	The electronics application needs analysis of characteristics of semiconductor devices to solve problems related to improve the gain.	Students will do the experiments to obtain the characteristic of semiconductor devices.	Students will be using Electronics virtual laboratory and gains proficiency on the same	Students follow basic ethical practice to prepare the assignment.

• Images / Screenshot of the practice:

Name: G. Ari Kara Moorthy Reg No: 953623103009 Application of diode:- 1) switches 2) Power supply 3) Rectifier 4) Digital System	S. Arida 953623103036 Application of diode. Switches Rectifier Power Supply Digital System
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• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that the students will be able to explain

the applications of diodes.

- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 90% of students answered.

Benefit of the practice:

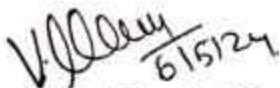
- Students were able to attend the question even in the questions are in indirect form.
- Students were able to explain the concepts in examination without any confusion.

❖ ***Challenges faced in implementation:***

- In the class mostly girls hesitate to answer the questions.
- Time utilization for conducting activity.

References:

- 1 John Bird, "Electrical Circuit theory and technology", Routledge; 2017.
2. Thomas L. Floyd, 'Electronic Devices', 10th Edition, Pearson Education, 2018.
3. Albert Malvino, David Bates, 'Electronic Principles, McGraw Hill Education; 7th edition, 2017
4. Muhammad H.Rashid, "Spice for Circuits and electronics", 4th Edition., Cengage India, 2019.
5. H.S. Kalsi, 'Electronic Instrumentation', Tata McGraw-Hill, New Delhi, 2010


6/5/24

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INSTRUMENTATION

Name of the Faculty member (s): Mrs.V.Krishna Meera

Innovative Practice Description

- Unit / Topic: Unit IV / Characteristics of BJT under CE configuration
- Course Outcome: CO4
- Topic Learning Outcome: TLO17.
- Activity Chosen: Virtual lab demonstration.
- Date: 09.05.2024
- Justification:

The structure, operation and V-I characteristics of transistor. After teaching this topic, I thought of this activity for making to have clarity about the operation of the transistor and its characteristics very easily. I can judge the understanding level of the students.

- Time Allotted for the Activity: 5 Minutes
- Details of the Implementation:

- After teaching the concept, I opened virtual lab software and explained how to give connections, how to obtain the characteristics of transistor in CE configuration. Finally, I taught to students how to draw the input and output characteristics of CE configuration.
- Total Strength :43
- Reporter: Myself
- At the end of the Class (Last 5 minutes)
- I asked the students to obtain the characteristics of transistor in CE configuration in virtual lab.
- Finally, I told the students to send the report of that experiment.

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO8	PO10	PO12
CI12.4	3	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

Form No. AC 10c

Rev.No.01

Effective Date: 02.08.2021

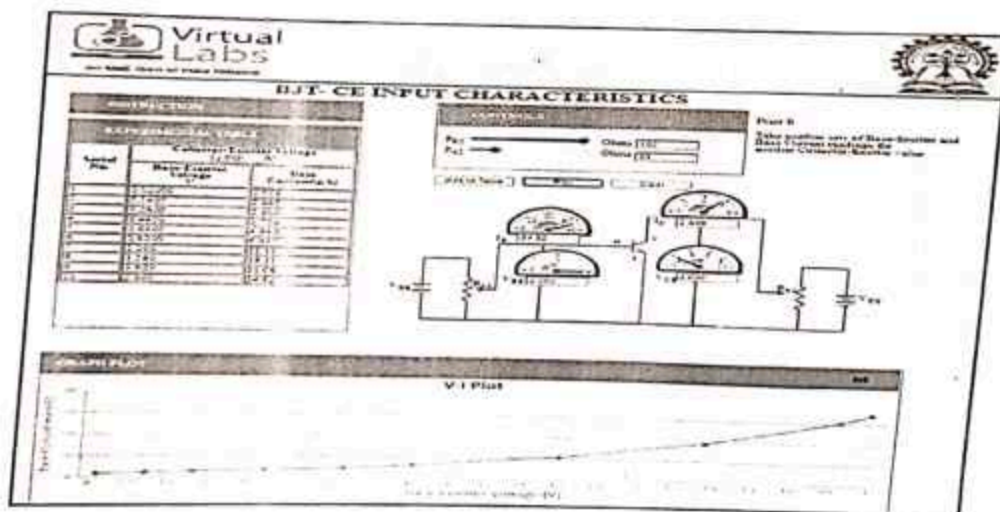


- PO / PSO mapped:

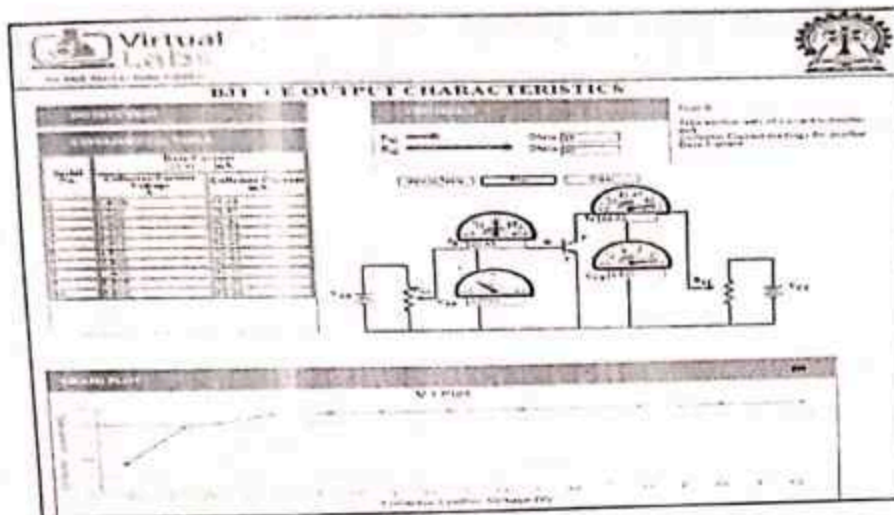
Innovative practice	PO1	PO2	PO3	PO8	PO10	PO12
	3	2	1	1	1	1
Justification for correlation	Students will be solving basic problem on Semiconductor devices using basic electrical concepts.	Suitable mathematical and engineering knowledge will be applied by the students to solve basic problems	The electronics application needs analysis of characteristics of semiconductor devices to solve problems related to improve the gain.	Students will do the experiments to obtain the characteristic of semiconductor devices.	Students will be using Electronics virtual laboratory and gains proficiency on the same	Presentation / Assignment will be submitted by the students on configuration..

- Images / Screenshot of the practice:

Input Characteristics of BJT under CE Configuration



Output Characteristics of BJT under CE Configuration



• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that the students will be able to understand the CE configuration of transistor.
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 85% of students answered.

Benefit of the practice:

- Students were able to attend the question even in the questions are in indirect form.
- Students were able to explain the concepts in examination without any confusion.

❖ Challenges faced in implementation:

- In the class some students hesitate to do this in virtual lab.
- Time utilization for conducting activity.

References:

1. https://www.nactateachers.org/images/The_Student-Developed_Quiz_or_Exam-Scaffolding_Higher-Order_Thinking.pdf
2. John Bird, "Electrical Circuit theory and technology", Routledge; 2017.
3. Thomas L. Floyd, 'Electronic Devices', 10th Edition, Pearson Education, 2018.
4. Albert Malvino, David Bates, 'Electronic Principles, McGraw Hill Education; 7th edition, 2017
5. Muhammad H.Rashid, "Spice for Circuits and electronics", 4th Edition., Cengage India, 2019.
6. H.S. Kalsi, 'Electronic Instrumentation', Tata McGraw-Hill, New Delhi, 2010.

V. K. Singh
9/15/24
Signature of Faculty Member

Form No. AC 10c

Rev.No.01

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Effective Date: 02.08.2021